



Pharm*Achieve*

CROUP

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

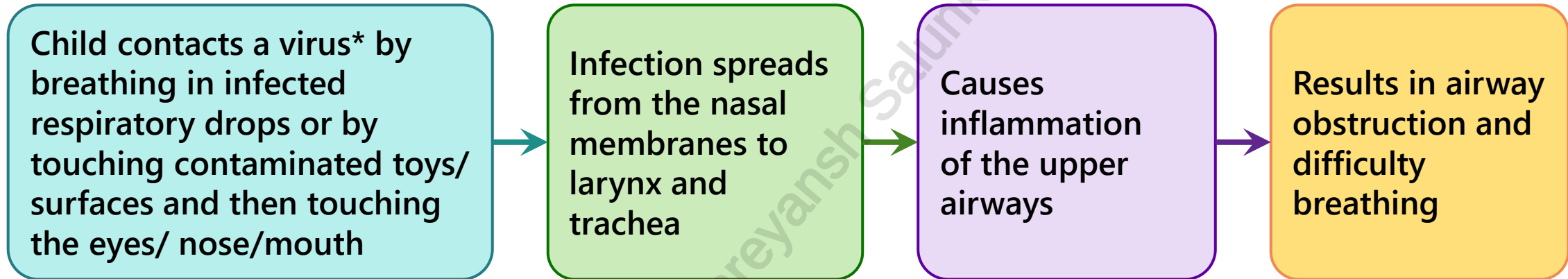
**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

WHAT IS CROUP?

Also known as laryngotracheobronchitis; an infection causing inflammation of the upper airways



*Viruses are the most common cause of croup, namely parainfluenza virus types 1 and 3. Other viruses that have also been isolated include: Influenza A and B, adenovirus, respiratory syncytial virus (RSV), metapneumovirus, coronavirus, and mycoplasma.

CLINICAL PRESENTATION

- Loud barking cough that worsens at night
- Noisy, laboured breathing
- Hoarse voice
- Possibly accompanied by a fever ($> 38^{\circ}\text{C}$ or 100.4°F)

RISK FACTORS

- Age (6 months - 3 years)
 - Although it may occur in children from 3 months old up to 15 years of age
- Gender (boys are affected more than girls)

DIAGNOSIS

Laboratory and radiologic examinations are NOT required

Patient history

- “Seal-like barky cough” may occur after child experiences a non-specific cough, rhinorrhea, and fever
- Symptoms worsen at night and improve during the day; aggravated by crying and agitation
- Symptoms of obstruction generally resolve within 2 days, however, a small number of children are symptomatic for up to 5-6 days

Physical examination

- Stridor: occurs during inspiration and if severe, may occur during expiration
- Chest wall retractions and respiratory distress: range from mild to severe and indicate the severity of the condition
- Respiratory failure occurs over a period of several hours; symptoms include lethargy, pallor, dusky appearance, cyanosis (blue-tinged skin), decreased breathing sounds, and reduced respiratory effort

Objective examination

- Lateral and anteroposterior (AP) soft tissue neck films (clarifies diagnosis in patients not responding to therapy or in those with atypical croup); pulse oximetry (in severe croup)

MEDICAL/DRUG RELATED CAUSES OF SIGNS AND SYMPTOMS

FYI

- Medications that cause a cough
- Allergic reaction
- Irritant cough (e.g. smoke inhalation)
- Asthma
- Aspiration or ingestion of a foreign body

GOALS OF THERAPY

- ✓ Decrease duration of croup symptoms
- ✓ Decrease severity of symptoms
- ✓ Minimize child and parent anxiety associated with croup
- ✓ Decrease risk of intubation
- ✓ Minimize hospitalizations and physician visits

NON-PHARMACOLOGICAL THERAPY

- **Maintain a calm and reassuring atmosphere**
- Oxygen administration through a plastic hose ("blow-by" oxygen)
- There is insufficient evidence for humidifiers, or helium-oxygen mixtures (i.e., heliox) – not routinely recommended
- Mist does not reduce respiratory distress, thus should not be used; placing children in mist can cause anxiety and agitation

The following symptoms are an indication of, or may lead to hypoxia and should be treated with oxygen in a hospital setting: difficulty breathing, wheezing or whistling sounds while breathing, blue-tinged skin

PHARMACOLOGICAL THERAPIES

Corticosteroids

Inhaled Adrenergic Agonists (Epinephrine)

Inhaled Beta-2-Agonists

Analgesics

Antitussives and Decongestants

Antibiotics

PHARMACOLOGICAL THERAPY: CORTICOSTEROIDS

Avoid in children with immunodeficiencies or who have been recently exposed to the varicella virus

- Corticosteroid therapy is the primary treatment for croup
- Proven to reduce the number of, need-for, and duration of intubations, number and duration of hospitalizations, need for additional inhaled epinephrine, and rate of return visits to medical professionals
- Oral dexamethasone or nebulized budesonide relieve symptoms 2–3 hours after administration
- Dexamethasone: 0.6 mg/kg PO as a single dose is usually administered, but may give IM/IV if persistent vomiting or respiratory distress
 - Mild to moderate cases: a lower dose (e.g. 0.15 mg/kg) may be used but it is unclear whether these smaller doses are equally effective
 - Severe cases: higher doses (0.6 mg/kg) could be more beneficial
- Nebulized budesonide is effective in treating croup and is equivalent to dexamethasone (PO), but its use is limited due to increased costs
 - Inhaled budesonide can be used in patients who experience sustained vomiting
 - There is NO benefit in administering inhaled budesonide in addition to systemic dexamethasone

PHARMACOLOGICAL THERAPY: INHALED ADRENERGIC AGONISTS (EPINEPHRINE)

- Administration of nebulized budesonide and epinephrine simultaneously, can be beneficial in those with severe croup or who are near respiratory failure
- Administration of nebulized epinephrine:
 - Substantially reduces the number of children with severe croup who require an artificial airway
 - Reduces respiratory distress within 10 minutes of administration and effects last for greater than 60 minutes (although effects of epinephrine wear off within two hours of administration)
 - Improves croup score in patients 30 minutes after treatment
 - Patients may be discharged 2-3 hours after epinephrine treatment as long as symptoms do not return
 - If multiple doses of nebulized epinephrine are needed, close cardiac monitoring should be performed, and a pediatric intensivist should be contacted

PHARMACOLOGICAL THERAPY: INHALED BETA-2 AGONISTS

- Beta-2-agonists (e.g. salbutamol) are **NOT used to treat croup**
 - No randomized controlled trials assessing their use in treating croup are available
 - They do not have an effect on the upper airways (the upper airways do not contain smooth muscle) thus their benefit in treating croup is unlikely

PHARMACOLOGICAL THERAPY: ANALGESICS, ANTITUSSIVES AND DECONGESTANTS, AND ANTIBIOTICS

Analgesics

- Analgesics can be used to reduce pain, agitation, and fever

Antitussives and decongestants

- Antitussives and decongestants are NOT used to treat croup
- No studies assessing their benefits are available

Antibiotics are NOT used to treat croup

- Croup is almost always a viral infection thus antibiotics cannot be used

PHARMACOLOGICAL THERAPIES: A QUICK REVIEW

Corticosteroids



Inhaled Adrenergic Agonists (Epinephrine)



Inhaled Beta-2-Agonists



Analgesics



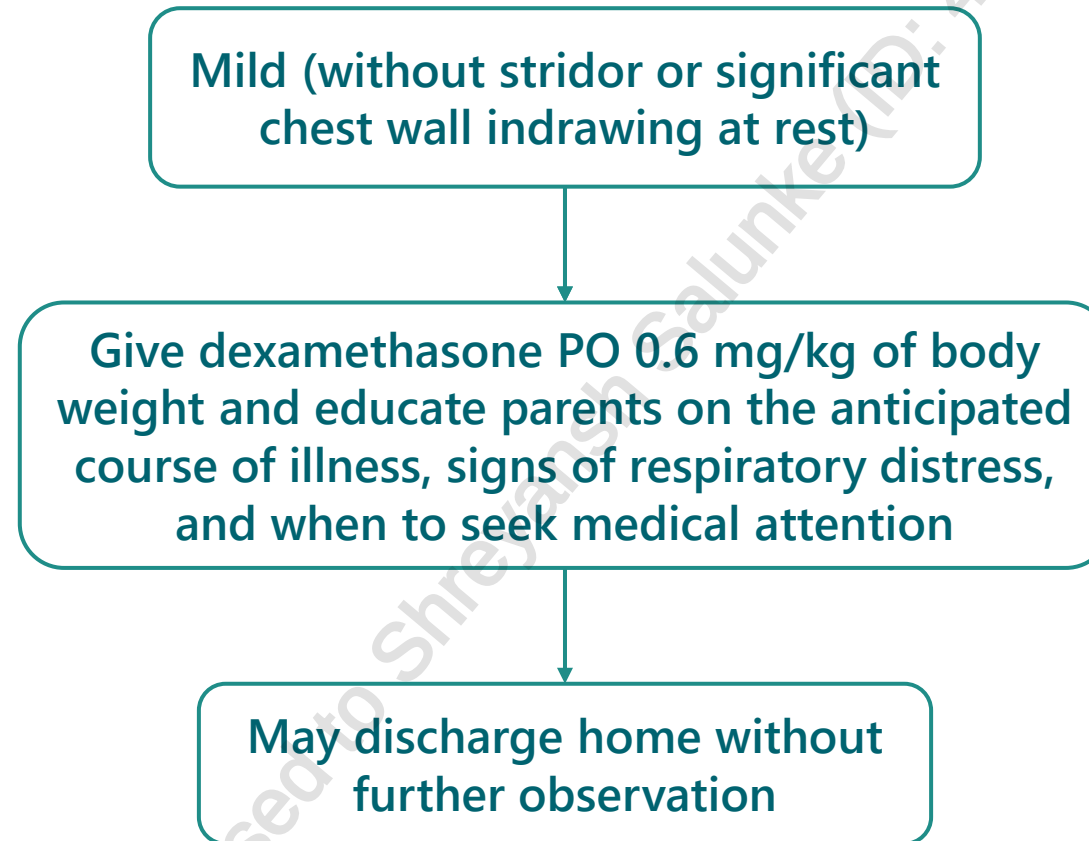
Antitussives and Decongestants



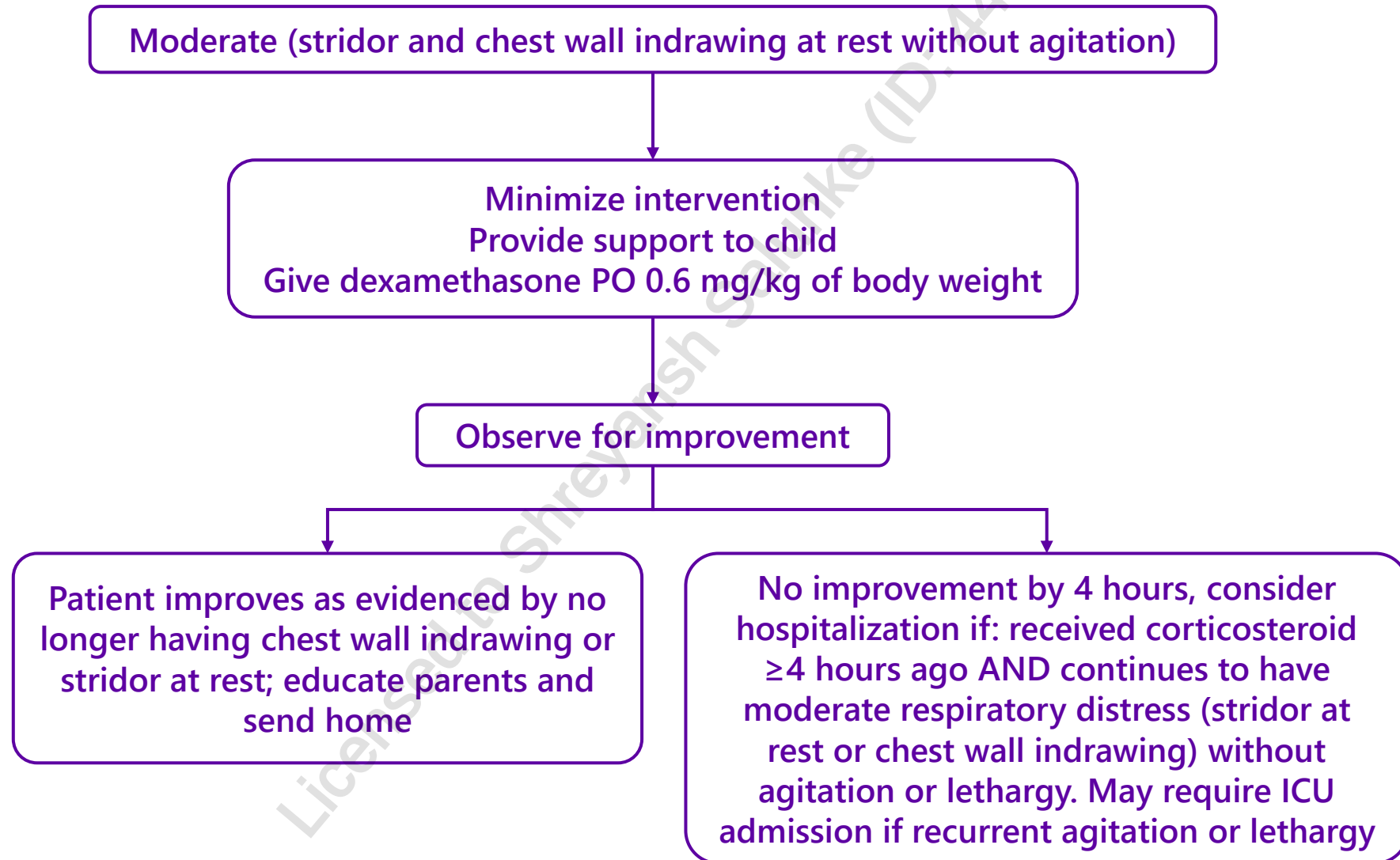
Antibiotics



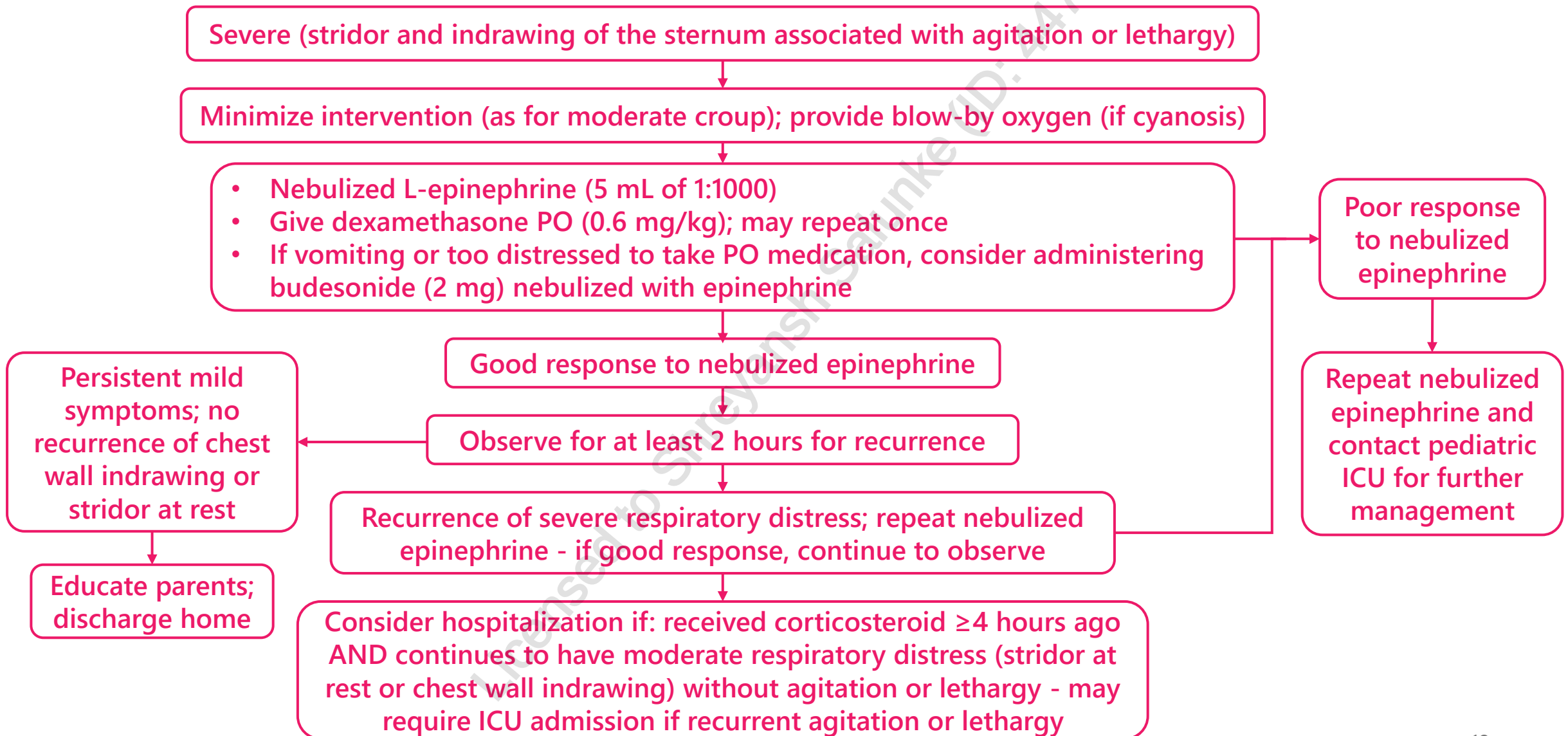
TREATMENT ALGORITHM FOR MILD CROUP IN THE OUTPATIENT SETTING



TREATMENT ALGORITHM FOR MODERATE CROUP IN THE OUTPATIENT SETTING



TREATMENT ALGORITHM FOR SEVERE CROUP IN THE OUTPATIENT SETTING



DOSING INFORMATION

Class	Drug	Dose	Comments
Adrenergic Agonists	L-epinephrine (Adrenalin)	5 mL of 1:1000 (1 mg/mL) solution via nebulizer	• Use for rapid response in children with severe respiratory distress; positive effects of epinephrine last up to 2 h • Patients should be observed for at least 2 h before discharge
Corticosteroids	Dexamethasone	0.6 mg/kg PO/IM/IV single dose May supplement initial dose in 6–24 h as necessary to a maximum daily dose of 0.6 mg/kg or 10 mg whichever is less	• Benefit demonstrated for all patients diagnosed with croup regardless of severity (mild-severe); reduces symptoms, sleep loss and parental anxiety, rate and duration of intubation, rate and duration of hospitalization and rate of return to medical care • While both oral and parenteral dexamethasone are effective, oral is preferred as it is less distressing for the child; onset of action: 2–3 h after treatment • No evidence to suggest multiple doses provide additional benefit over a single dose
Corticosteroids	Budesonide (Pulmicort Nebuamp)	2 mg (4 mL of 0.5 mg/mL solution) via nebulizer May repeat dose in 6–24 h if necessary	• Inhaled budesonide has been shown in several studies to be equivalent to oral dexamethasone but is more expensive • A good option for patients with vomiting or severe respiratory distress (may be nebulized together with epinephrine)

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Croup symptoms (e.g. noisy/laboured breathing)	Improve	2-3 h after corticosteroid therapy
Obstructive symptoms	Resolve	48 hours (symptoms persist for up to 5-6 days in a small number of children)

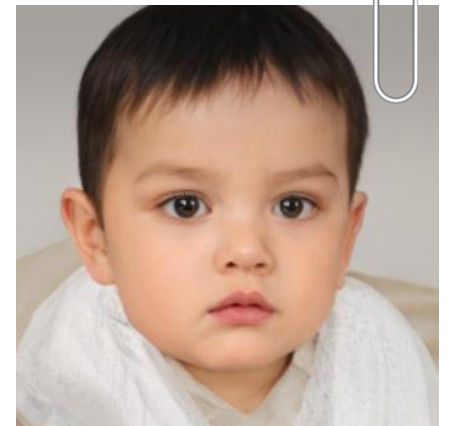
TAKEAWAYS

- Breathing in cool air may provide relief to children with mild to moderate croup: they should dress warmly and open a window or freezer door for a couple of minutes to breathe in cool air
- The preferred corticosteroid treatment is dexamethasone (PO) 0.6 mg/kg as a single dose
 - If the child is vomiting, extremely ill or is physically unable to take oral dexamethasone, IM/IV dexamethasone OR nebulized budesonide should be considered
 - Nebulized budesonide should be considered instead of IM or IV dexamethasone as parenteral administration will be upsetting for a child who is already distressed

CASE

JU is a 20-month-old male brought into the emergency department of the hospital by his parents in the evening with complaints of a low-grade fever, hoarse voice, and a barking cough that worsens at night. His parents explain that these symptoms have occurred over the past 48 hours. They report noisy breathing and worry that JU is experiencing dyspnea whilst crying. JU has been waking up in the middle of the night, distressed with labored breathing, and he has been less interested in taking fluids. During a physical examination by his medical team, JU has inspiratory stridor at rest, a temperature of 38.3°C, moderate intercostal retractions, and an oxygen saturation of 93% on room air. He is lethargic and irritable, and he has a blue discoloration to his face. He shows no signs of vomiting. His parents confirm that JU's vaccinations are up-to-date, and he has no known drug allergies. His past medical history is unremarkable. JU weighs 11.5 kg.

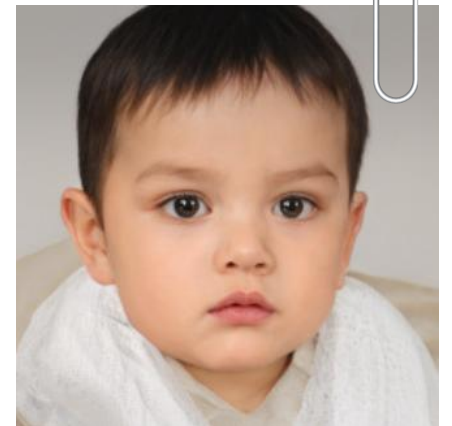
1. What is the role of blow-by oxygen in JU's management?
 - a) It should be avoided as it worsens agitation
 - b) It improves bronchodilation when combined with pharmacotherapy
 - c) It replaces the need for pharmacotherapy in mild cases of croup
 - d) It is preferred over an oxygen mask in a distressed child



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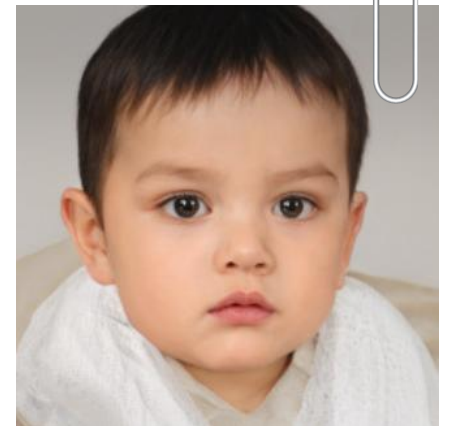
2. What is the most appropriate pharmacologic therapy to initiate in JU?
- a) Dexamethasone 1.7 mg PO once
 - b) Nebulized 5 mL of L-epinephrine 1:1000
 - c) Nebulized budesonide 2 mg
 - d) Dexamethasone 6.9 mg IM followed by a repeat dose after 6 hours



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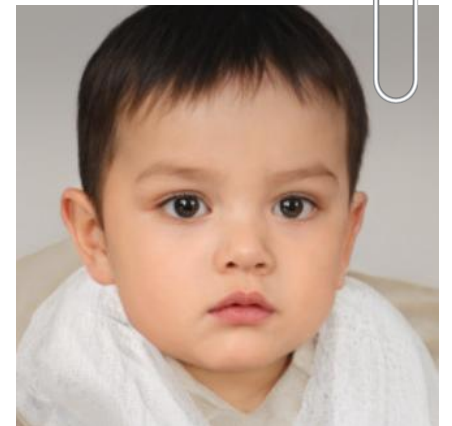
3. How long should JU be observed for following pharmacotherapy?
- a) 1 hour
 - b) 2 hours
 - c) 4 hours
 - d) 6 hours



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4. Which of the following statements should be included when counselling JU's parents at discharge?
- a) Avoid exposure to cold air as it can worsen symptoms
 - b) Administer dexamethasone immediately if JU's fever returns
 - c) Administer a salbutamol inhaler with a spacer device if JU's dyspnea symptoms persist
 - d) Return to the emergency department if JU appears distressed with high-pitched noisy breathing



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CHANGE LOG

July 2025

- Slides 22, 23: Case study and questions added
- Formatting changes made; added 'FYI' to some slides

October 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Minor updates to non-pharmacological interventions
- Slide 11: Minor update to dexamethasone dosing